

NWFP Group Management and Marketing Plan



Name of NWFP Group:

Village: Lamtang

Gewog: Goshing

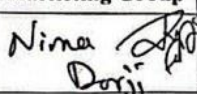
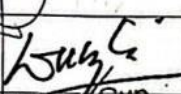
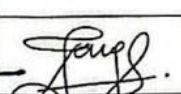
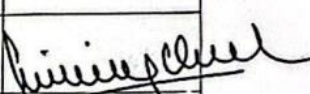
Dzongkhag: Zhemgang

Plan Period: 2024 –2034 (*1st Revision*)

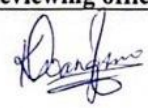
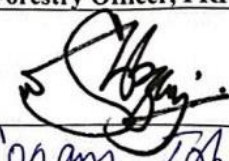
Approval sheet

Name of NWFP Management and Marketing group	Lamtang NWFP Management and Marketing Group
Village(s)	Lamtang
Gewog	Goshing
Dzongkhag	Zhemgang
No. of hh	20
Area (ha)	42 ha
Plan period	2024-2034

Prepared and submitted for Approval

	Chairperson of NWFP Management and Marketing Group	Gup of the Gewog	Head FRPMS of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
Name:	Nima Dorji	Namgyal	Jangchu Wangdi	Tashi Wangchuk
Date:	31/3/24	Goshing Gewog Panbang Zhemgang		

Reviewed and recommended for Approval

	Reviewing officer, FRPMD	Chief Forestry Officer, FRPMD
Signature		
Name	Keang Wangmo	Sonam Tobgay
Date	24-7-24	25/07/24

Approved by:

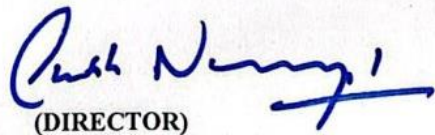

(DIRECTOR)
Department of Forests and Park Services

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Acronyms

AHL	Annual Harvesting Limit
BDBL	Bhutan Development Bank Limited
DFO	Divisional Forest Office
Hh	Household
ITMS	Institute of Traditional Medicine Services
Kg	Kilogram
Km	Kilometer
Ac	Acre
Ha	Hectare
KNC	Khenrig Namsum Cooperatives
NWFP	Non-wood Forest Product
FRPMD	Forest Resources Planning and Management Division
FRPMS	Forest Resources Planning and Management Section

Part A: NWFP Management and Marketing Plan

1. Background

Lamtang Non-wood Forest Product (NWFP) Management and marketing Group is located under Goshing Gewog, Zhemgang Dzongkhag. It is about 15 minutes-drive from Goshing Gewog Center. The community has access to modern amenities like Basic Health Unit (BHU), primary school, Renewable Natural Resources Extension Centre (RNR-EC), forest office, electricity, water supply, motorable road and telecommunication network. Agriculture is the primary source of livelihood in the village and people cultivate maize, rice, potatoes, beans, cabbage, chilly and cash crops like orange, ginger and cardamom. They also depend on forest for timber, firewood and NWFPs for additional income generation for their livelihood. The village has 20 households who took interest on NWFPs management and all are registered member of Lamtang NWFP management and marketing group. There are various NWFPs found in their locality such as *Neomicrocalamus andropogonifolius* (Yula), *Diplazium esculentum* (Jeema), *Knema tenuinervia* (Durka), *Borinda grossa* (Baa), *Piper betleoides* (Dhem pan), *Oroxylum indicum* (Namkaling), *Thysanolaena maxima* (Tsakusha), *Tupistra nutans* (Ngawa pem/Thorma Ju), *Embllica officinalis* and *Dendrocalamus hamiltonii*. Though community harvests all those species for self-consumption and few for commercial purposes. They have prioritized *Dendrocalamus hamiltonii* for commercialization. There are five bamboo management areas consisting 42 ha. which will be management by community during the plan period.

Throughout the earlier management plan, community couldn't conduct the activities as per plan due to Covid-19 pandemic. But the group has marketed the bamboo shoot. This is the 1st revision of management plan and the previous plan had expired in June, 2022. Before reviewing the management plan, a comprehensive consultation meeting with the management group members was conducted to understand their inactiveness and to assess their interest in continuing further management. Capturing their interest in continued management of NWFPs, Divisional Forest Office (DFO) Zhemgang provided technical support to the group in resource assessment and preparation of management plan. The management plan will guide the group members in proper harvesting technique for sustainability of the resources.

2. Goal and objectives

1. To increase the income through sale of resources
2. To utilize the resources on sustainable basis
3. To share the resources on equitable basis amongst the members

3. Description of NWFPs present in the area demarcated for management.

3.1 *Dendrocalamus hamiltonii*: It is a nice bamboo with straight strong (near solid) culms and edible shoots. It has very tight clumping growth habit and slightly weeping tops with lush foliage. Some of the culms may have a cream striation on the lower inter-nodes. Culms is 9-25cm in diameter, strong pendulous above, densely covered in persistent, brown and white, furry and wax; wall thin; branches fewer toward base, center branch to 5cm in diameter, smallest branches curving from clump, nodes with dense, long aerial roots. The group will collect the resources from Dangling 1, Dangling 2, Wangling, Kodong 1 and Kodong 2 blocks. It is 30 minutes to 1 hour drive from the village.

3.2 *Neomicrocalamus andropogonifolius*: It is a perennial plant with the following characteristics: pluricaespitose, elongated rhizomes, scandent culms, it grows up to 1200cm long, woody, culm-internodes terete, thin-walled, culm nodes flush with internodes, leaf-blade base broadly rounded, leaf-blade margins scabrous and leaf-blade apex acuminate. It is mainly used for crafting purposes. Management group wanted to supply the resources to Bjoka Tsharzo Gongphel Tshogpa for bamboo crafting and also wanted to train themselves for crafting of bamboo products. But due to flowering of Yula, the assessment couldn't be conducted.

3.3 Diplazium esculentum: Is a straggling fern that grows to 1m tall. The young shoots or fronds of the fern are consumed as vegetables, sold in local markets throughout Asia and the Pacific Islands. Fronds are variable in size, bipinnate, linear leaflets with lobed to serrated margins. Is the vegetable fern, edible fern found throughout Asia and Oceania. It is probably the most commonly consumed fern. The management group would collect the resources within their managed area and wanted to sell fresh or processed fern in the market.

3.4 Knema tenuinervia: Is an evergreen shrub or tree, usually growing from 3 - 20 meters tall with some specimens reported to reach 30 meters. The tree is harvested for its wood, which is used locally and especially for processing into traditional medicines.

3.5 Borinda grossa: Is an evergreen Bamboo growing to 10 m (32ft) by 3 m (9ft). The flowers are hermaphrodite (have both male and female organs) and are pollinated by Wind. Suitable for: light (sandy), medium (loamy) and heavy (clay) soils and prefers well-drained soil. Suitable pH: acid, neutral and basic (alkaline) soils. It can grow in semi-shade (light woodland). It prefers moist soil.

3.6 Piper betleoides: The betel plants are an evergreen perennial, with glossy heart-shape leaves and white catkin. The betel leaf plant is a creeper grown in our country too and belongs to pepper family. Its waxy green, heart-shaped leaves are used for medicinal and culinary purposes.

3.7 Oroxylum indicum: Is a monotypic species of flowering plant belonging to the genus Oroxylum and the family Bignoniaceae, commonly called midnight horror, oroxylum, or Indian trumpet flower. It is a mesocaul tree which can reach a height of 18 meters. The resources are also presence within the managed NWFP area of the group.

3.8 Thysanolaena maxima: A strongly tufted, very robust perennial grass with erect or slightly spreading solid bamboo-like culms up to 3.5 m tall. Leaf-sheath hairy along outer margin; ligule a scarious membrane, 1-2 mm long; leaf-blade lanceolate-acuminate, 30-65 cm x 3-7.5 cm, base broad and rounded or subcordate, margins scaberulous, conspicuously glaucous beneath. Inflorescence a terminal huge and drooping panicle, up to 140 cm long, well exerted, branches divided and subdivided into many branchlets; spikelets awnless, short pedicelled, falling with part of the pedicel, often in pairs on a common peduncle, 2-flowered; lower glumes clasping. Caryopsis subglobose to ovoid, 0.6 mm long, reddish-brown. Members wanted to collect the resources markets after processing.

3.9 Tupistra nutans is an amazing Indian native that makes a nice clump resembling an aspidistra on steroids. Each clump is composed of 3' long narrow dark green leaves that are held upright. The flower of species is used as vegetable which is sold expensively in the market.

3.10 Phyllanthus emblica: *Phyllanthus emblica* Gatertn, commonly called amla, has traditionally been used for different medicinal purposes including: rheumatic pains, gonorrhoea, asthma, haemorrhage, jaundice, dyspepsia, nausea, constipation, diarrhoea and eye disease.

Table 1. Brief description of NWFPs for commercial harvesting in each season

NWFP name	Harvesting area(s) (name)	Parts harvested	Use/purpose (traditional & others)	Harvesting season (months)	Estimated quantity for harvest in a season/year
<i>Neomicrocalamus andropogonifolius</i>	Laptang	Culms	Crafting	December-May	2500bundle
<i>Dendrocalamus hamiltonii</i>	Zomphu, Pangling, Maja and Dramlang	Shoots	Vegetables	May-November	5000kg
		Culm	Construction /crafting	Whole year	3500 piece
<i>Diplazium esculentum</i> (fern)	Kodong and Maja	Tender shoot	vegetables	April-September	1000 bundles

	Pangling				
<i>Knema tenuinervia</i>	Pangling	Timber	For religious	Whole year	1000kg
<i>Borinda grossa</i>	Lamtang	Culm	Crafting	Whole year	3500pieces
<i>Piper betleoides</i>	Lamtang	Leaf	consumption	Whole year	500 bundles
<i>Oroxylum indicum</i>	Lamtang	Upper parts, inflorescence	For religious	August-November	2500 pieces
<i>Thysonolaena latifolia</i>	Khawangbe	Leaf/ inflorescence	Fodder/soft broom	December to March	1500 bundles
<i>Cinnamomum tamala</i>	Lamtang	Leaf	Stimulant/tea	Whole year	1000 bundles
<i>Tupistra nutans</i>	Kodong & Maja Pangling	Flower	Vegetables	September-November	200kg
<i>Phyllanthus emblica</i>	Kodong & Maja Pangling	Fruits	Fruits	August-January	500kg

4. Procedures for sustainable harvesting of NWFPs

The NWFP management group members have to adhere to specific procedure for the sustainable management of resources.

- The Chairperson will consult with forest office and obtain permit for collection during the onset of harvesting season.
- The Chairperson will announce starting date for harvesting only after obtaining required permits from forest office.
- Only one member from each household will be entitled for collection of NWFP products.
- Collection will be done within the demarcated area only.
- Harvesting prescriptions will be strictly followed by the collectors (Table 2).
- The Vice-Chairperson will maintain proper record of harvest and sale of NWFPs for monitoring and evaluation.
- Chairperson will notify the forest office on completion of harvesting the NWFPs.
- Treasurer of NWFPs management group to maintain all financial records

Table 2. Prescriptions for sustainable harvesting of specified NWFPs

NWFP Species	Harvesting prescription
<i>Neomicrocalamus andropogonifolius</i>	▪ Cut only feasible culm ▪ Do not cut all culm & retain at least two - Harvest culm when it produces secondary shoot
<i>Dendrocalamus hamiltonii</i>	• Collect shoot up to 25% from the clump • Use sharp knife to cut the shoot without damaging other parts in the stand • Cover the cut portion of the shoot that is left in the stand with soil
<i>Diplazium esculentum</i> (fern)	❖ Collect the fern that is above the ground level only ❖ Do not up root the whole plants ❖ Just break the base of the fern to collect the shoot
<i>Knema tenuinervia</i>	✓ Collect only fallen trees ✓ Never cut standing trees
<i>Borinda grossa</i>	➤ Collect matured culm ➤ Do not disturb the young shoot ➤ Harvest the culm from the base to maintain sanitation
<i>Piper betleoides</i>	❖ Collect green leave ❖ Do not cut down the trees ❖ Do not break down the stem

	❖ Climb up the trees to collect the leaves seeing the possibilities
<i>Oroxylum indicum</i>	• Collect flower when matured • Never cut down the trees • Handle with care while harvesting the flower
<i>Thysonolaenna latifolia</i>	➤ Collect the matured parts/stem ➤ Never cut it from the base ➤ Do not uproot the stem
<i>Phyllanthus emblica</i>	○ Collect when future ○ Do not cut down /bend the trees
<i>Tupistra nutans</i>	❖ Just collect flower only ❖ Never uproot the stem

5. Block wise area and distances for prioritized species

There are various collection areas for different NWFPs that will be managed by Lamtang NWFPs management group but very minimal. The areas and walking distances for prioritized species especially for *Dendrocalamus hamiltonii* is as tabulated in Table 3.

Table 3. Details of collection area

Village	Collection area	Distance from village	Total area(ha)
Marangduth	Dangling 1	1.3hrs	11/2
	Dangling 2	1hr	8
	Wangling	1.40hrs	15
	Kodong 1	1-2hrs	4
	Kodong 2	1-2hrs	3.8
Total			42ha (103.78acres)

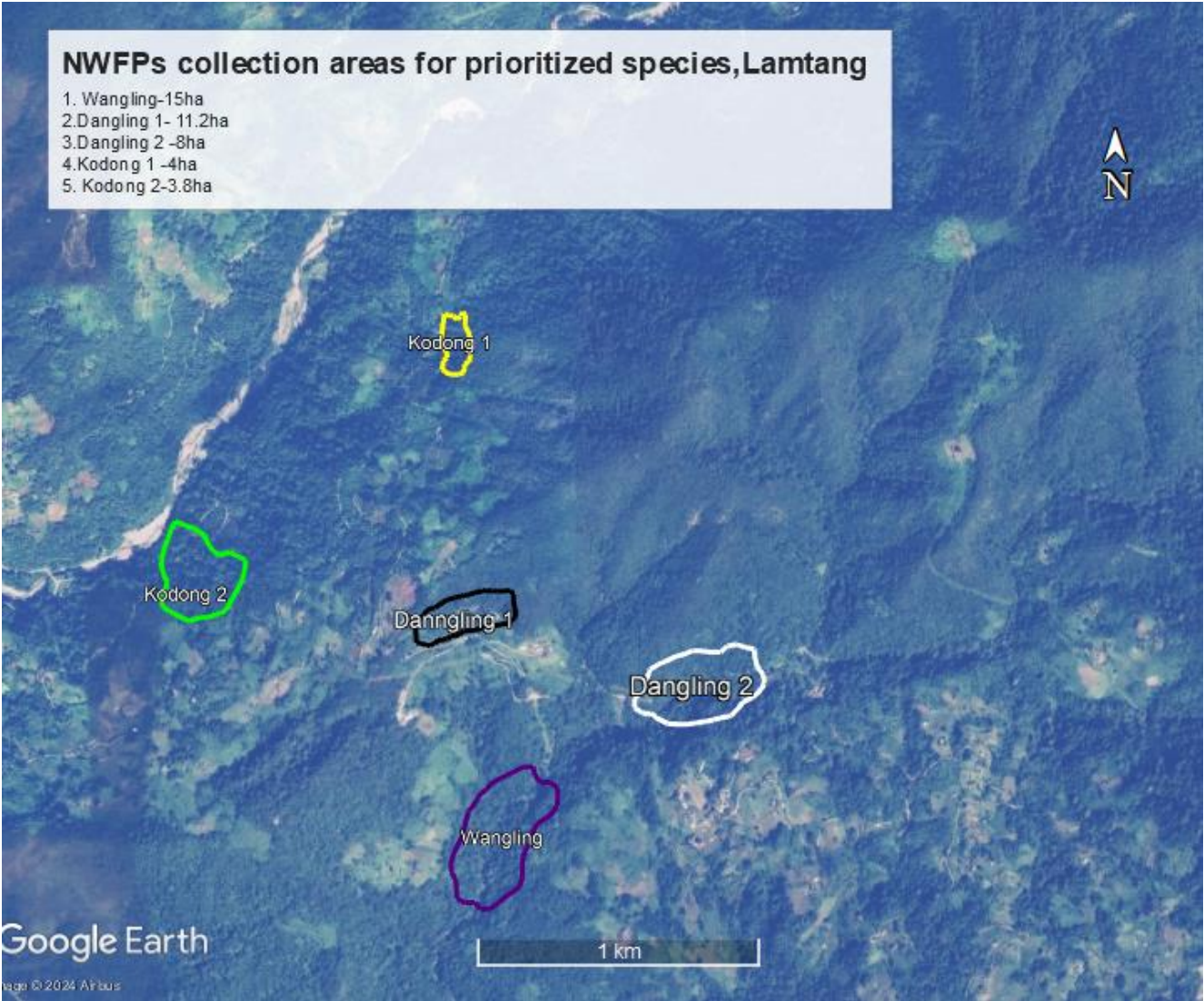


Figure 1. Location map of the NWFP harvesting areas

6. Forest Resource Assessment for Dendrocalamus hamiltonii

AHL= 33% of the total stock of the 1+year old culms +2+year old culms+ 70% of 3+years old culms
In which the 33% of the stock of the 1+year old and the 2+years old culms should not exceed the total of the harvestable culms of the combined age classes.
* As calculated as per productive area in sub-block/block (calculate the average per sample plot and divide by 0,025(10*25)/10000) to get the number of culms

Per ha, multiply with the productive area(ha) of sub-block/block to get the total of culms per sub-block/block: compare the result with the culms considered harvestable by the villagers (calculated in the same way).

Table 4. Dendrocalamus hamiltonii resource assessment compilation form

Resource assessment compilation form for Dendrocalamus hamiltonii							
Block name:	All blocks below		Dates of bamboo resource assessment:		05-02-24		
Sub-block (number)	Name of sub-block		Area (ha)	% of productive area	Number of sample plots		
Wangling	Wangling		15	85	16		
Dangling 1	Dangling 1		11.2	92	11		
Dangling 2	Dangling 2		8	96	9		
Kodong 1	Kodong 1		4	98	7		
Kodong2	Kodong2		3.8	88	7		
			42				
Age class	Number of culms	Block 1(Wangling)	Block2 (Dangling1)	Block 3(Dangling2)	Block4 (Kodong 1)	Block5 (Kodong2)	Total Culms
1+ year old bamboo culms	Total counted in sample plots	90	59	57	33	31	270
	Total harvestable in sample plots	65	54	43	27	30	219
	Total in block*	3600	2360	2280	1320	1240	10800
	Total Harvestable in block*	2600	2160	1720	1080	1200	8760
2+years old bamboo culms	Total counted in sample plots	79	52	52	32	39	254
	Total harvestable in sample plots	65	45	46	28	25	209
	Total in block*	3160	2080	2080	1280	1560	10160
	Total Harvestable in block*	2600	1800	1840	1120	1000	8360
3+years old bamboo culms	Total counted in sample plots	82	62	56	32	33	265
	Total harvestable in sample plots	71	53	43	24	28	219
	Total in block*	3280	2480	2240	1280	1320	10600
	Total Harvestable in block*	2840	2120	1720	960	1040	8680
Total		18080	13000	11880	7040	7360	57360
AHL		5965	5992	5385	3152	3302	23796

7. Demand and annual harvesting limit for prioritized Bamboo products.

Table below denotes the demand of community members for Dendrocalamus hamiltonii shoot and culms and its annual harvesting limits.

Table 5. Demand and AHL comparison

Total Annual Demand	Annual Harvesting Limit	Variation	Remarks
5000 pieces culms	23796 pieces of culms	18796 pieces of culms	Surplus

8. Marketing and sales plan

NWFPs collected by Lamtang NWFP management and marketing group have planned to market and sell products to listed potential markets:

- 1. National Institute of Traditional Medicine Service (NITMS)
- 2. Food Cooperation of Bhutan (FCB) which will be operated as farmers shop at Gewog level
- 3. Trader who deals with the resources
- 4. Khengrig Namsum Cooperative at Tingtibi, Zhemgang
- 5. Local Markets at Panbang
- 6. Nearby border towns
- 7. Thimphu CSO market
- 8. Yula would be sold to Bjoka Tsharzo Gongphel Tshogpa/Individual Bjoka people

The quantity, price of products and the value of expected sales will be done as per the data listed in table 4.

Table 6. Expected sales of product

NWFP	Potential Market	Estimated quantity for sale per season (unit)	Expected price per unit	Total (Nu)

<i>Neomicrocalamus andropogonifolius</i>	Bjoka Tsharzo Gongphel Tshogpa	2500bundle	350/bundle	875000
<i>Dendrocalamus hamiltonii</i>	KNC and construction sites	5000kg	12/kg	60000
		3500 piece	15/piece	52500
<i>Diplazium esculentum</i> (fern)	Khengrig Namsum Cooperative at Tingtibi, Zhemgang	700bundle	15/bundle	10500
<i>Knema tenuinervia</i>	Local Markets at Panbang	1000kg	75/kg	75000
<i>Borinda grossa</i>	ITMS, border towns and intruder	500 pieces	5/piece	2500
<i>Piper betleoides</i>	Border towns, trader and Local Markets at Panbang	500kg	300/kg	150000
<i>Oroxylum indicum</i>	Border towns, trader and Local Markets at Panbang	1500 pieces	15/piece	22500
<i>Thysonolaenna latifolia</i>	Border towns, trader and Local Markets at Panbang	1000 bundles	20/bundle	20000
<i>Cinnamomum tamala</i>	Border towns, trader and Local Markets at Panbang	200 bundles	30/bundle	6000
<i>Tupistra nutans</i>	Flower	200kg	150/kg	30000
<i>Emblica officinalis</i>	Fruits	300kg	150/kg	45000
Total Amount projected				1349000

8.1 Marketing & Sales plan for prioritized bamboo products

The community of Lamtang NWFP group has also projected the sales plan for prioritized *Dendrocalamus hamiltonii* resources mentioned table 5.

Table 7. Detail of sales plan

NWFP	Products (Local Name)	Unit	Rate per kg/Piece
<i>Dendrocalamus hamiltonii</i>	Boil Bamboo shoot	Kg	25/-
	Bamboo pickle	Bottle 400gm	230/ bottle
	Bamboo pickle	Bottle 200gm	130/-
	Dry bamboo shoot	200gm	130/-pkt
	Preserve bamboo shoot	Kg	100/-kg
	Preserve bamboo dry shoot	Dray	300/-dray
	Tangloop	Piece	450/-piece
	Basket	Piece	800/-piece
	Bamboo hat	Piece	500/-piece
	Roong	Piece	1000/piece
	Zortay	Piece	100/-piece
	Shera	Piece	350/-piece

8.2 Seasonal calendar for NWFP management group

The table 8 indicates the seasonal calendar for collection of NWFPs by the management group.

Table 8. Seasonal calendar for collection of NWFPs

Program	Bhutanese Calendar											
	1	2	3	4	5	6	7	8	9	10	11	12
Collection of Bamboo shoots & processing												
<i>Neomicrocalamus andropogonifolius</i>												

collection and trading												
<i>Diplazium esculentum</i> (fern) collection & marketing												
<i>Knema tenuinervia</i> collection & marketing												
<i>Borinda grossa</i> collection & marketing												
<i>Piper betleoids</i> collection & marketing												
<i>Oroxylum indicum</i> collection & marketing												
Plantation activities for selected species												
Weeding & cleaning activities for all species												
Collection & marketing of <i>Thysonolaenna latifolia</i>												
Collection & marketing of <i>Cinnamomum tamala</i>												

9. Marketing Arrangement

Following are the marketing strategies for the Lamtang NWFP group.

1. Products will be collected & transported up to the road point by the members.
2. Products like cane shoot, mushroom, damroo, and ferns will be sold in local market.
3. Piper will be sold to nearby Indian town and other market places.
4. Yula to local markets and finished products at Thimphu and other towns.
5. Cane to Bjoka Tsharzo Gongphel Tshogpa and other traders.
6. *Dendrocalmus hamiltonii* to KNC, Thimphu and other areas.

9.1 Procedures for NWFP harvesting, marketing and income sharing

Activity	Brief description of how the activities are carried out
Harvesting of products	• The Chairperson of NWFP group will obtain harvesting/collection permit from forest office in writing. • The Chairperson will announce starting date for harvesting only after obtaining required permits from forest office. • Only one member from each household will be collect NWFP for the first week and after that no limits is set for member involvement for collection. • Harvesting prescriptions will be strictly followed by the collectors. • Chairperson shall inform Range Office of completion of harvest.
Post-harvest practices	• The collectors are responsible for post-harvest processing of NWFPs.
Marketing of products	• The NWFPs will be handed over to the executive members for marketing after post-harvest processing. • The executive members will sell the produce in attendance of the group members to maintain transparency. In future the group plans to construct a common sale counter at the center to facilitate marketing of the NWFPs products.
Identification of buyers	• The Chairperson supported by other executive members will monitor and explore the demand from nearby institutes, agents, companies and other business groups. • Support will also be sought from Department of Agriculture and Marketing Cooperatives (DAMC)
Price fixing	• The executive members will coordinate.
Income sharing	• The income generated from the sale of NWFPs will directly share among the members. All the members will have equal right for sustainable harvesting of products as per the entitled lists in the plan. The group will also utilize the common fund during the community rituals and natural calamities with basic understanding among the members
Raising of community development	• A joint account in the name of the group will be opened under BDBL Branch Office located at Panbang. The account shall be managed by the committee members on behalf of the group.

fund	Annual membership fees, penalties and certain amount of sale proceeds will be deposited in the community development fund.
Payment to committee members for their service	5% of total income from the sale of products will be deposited into group's account and 15% from the sale will be shared within the committee members as work charges.

10. Benefit to the members

The income generated from the sale of NWFPs will directly share among the members. All the members will have equal right for sustainable harvesting of products as per the entitled lists in the plan. The group will also utilize the common fund during the community rituals and natural calamities with basic understanding among the members.

10.1 Financial projection for prioritized species and benefit to the members

Table 9. Financial projection for prioritized NWFPs

Sl. No	Particulars	Products (names)				Total Amount (Nu)
		Dendrocalamus hamiltonii		Tupistra nutans	Emblica officinalis	
		Shoot	Culm			
1	Revenues					
A	Total quantity (unit)	5000kg	3500pieces	200kg	300kg	
B	Market price (Nu per unit)	12	20	150	150	
C	Total revenues from forest product sales (a*b)	60000	70000	30000	45000	205000
A	Total revenues					205000
2	Expenses					
A	Packaging materials	-		-	-	-
B	Royalty	3500pieces@12			-	42000
C	Transportation costs					40000
D	Compensation fee to executive committee for marketing products			-	-	-
E	Contribution to community development fund	5% of total revenue		-	-	10250
F	Incidental expenses	-		-	-	-
B	Total expenses					90250
C	Total benefit (net-income) of group members (Total revenues A – Total expenses B)					114750
D	Average income to group members (C/ no of group members)					5737.50
E	Price per unit paid to group members (C/1a)					12.75

10.2 Financial projection of the benefits to the group, the community development fund

Table 10. Financial projection for community development fund

Sl. No	Particulars	Amount (Nu)
	Revenues	
1	Membership registration fees (20hh @100)	2000
2	Annual contribution	10250
A	Total revenues	12250
	Expenses (only include expenses that will be paid from the group funds)	-
		
B	Total Expenses	-
C	Balance: Total revenues – Total expenses	12250

Part B: Group Regulation and Norms (By-Law)

NWFP management & marketing group	
Membership arrangement and resignation	Any household with census recorded in Lamtang village and residing in the area can join the group. However, acceptance in the group will be made only by the group members through assembly & submitting letter of interest to the committee members.
Responsibilities and rights of members:	<u>Responsibilities and rights of members:</u> • Implement annual activities that are reflected in the management plan. • Manage the area in adherence to the management plan and refrain from doing illegal activities. • Attend ad-hoc/ planned meetings or activities for development of the group. • Attend any training related to NWFP. • Choose or reject committee members for better management of the group. <u>Procedure for becoming new members:</u> • Any household wishing to become members in the group shall submit application to the Chairperson. • The applicant should be 18 years of age to become a member. • The application shall be put forward in the general meeting for discussion and every member has right to accept or reject the application. • Membership fee of Nu.2000/- shall be levied upon registration. <u>Procedure for leaving the group:</u> • Any member wishing to discontinue their membership has right to do so. However, he/she should submit application to the Chairperson. • He/she can leave the group after discussing with the committee members. • Member leaving the group will not be allowed to avail their benefits from the group fund. • If their children wish to join as a member in future, they have to follow the procedure set to become new member.
Executive committee of NWFP management and marketing group	
Chairperson	Name of Chairperson: Nima Dorji Contact No. 17968041 <u>Responsibilities of Chairperson:</u> • Chair all the group meetings in which the group will have half yearly meeting twice in a year & a general meeting • Explore market for NWFPs. • Obtain collection permit from forest office during the onset of harvesting season. • Coordinate and plan activities for the user group along with the members and other stakeholders. • Coordinate and conduct meetings with members for development of the group. • Resolve conflict among group members. • Submit annual progress report to forest office. • Maintain record of all the group activities.
Secretary	Name of Secretary: Rinchen Cheten Contact No. 17301588 <u>Responsibilities of Secretary:</u> • Officiate in absence of Chairperson. • Assist Chairperson in coordinating and planning group activities. • Explore market for NWFPs.

	• Maintain records of any activities conducted by the group. • Maintain attendance record while conducting meeting and group activities. • Record minutes of meeting.
Treasurer	Name of Treasurer: Sangay Karma Contact No. 17392159 <u>Responsibilities of Treasurer:</u> • Officiate in absence of Chairperson and Secretary. • Collect annual membership fees, fines, amount from sale proceeds and deposit in the group account. • Maintain proper financial records in favor of the group. • Present financial status/ progress report during general meeting. • Assist Chairperson and Secretary in coordinating and planning group activities.
Tenure of committee member	• Their tenure shall be till the plan period ends. • However, if a committee member does not fulfill his/her responsibilities, misbehave and threaten the group's network, he/she shall be immediately replaced and be deprived of such opportunity in future.
Incentives to the committee members	• If the committee members have worked positively towards betterment of group, he/she shall be re-elected. This is to reward the individual for his/her dedicated service to the group.
Sources income generation for group fund	
Membership registration fees. Annual contribution Collection from sale proceeds	• Membership fees of Nu.100 per member household shall be collected annually. • The amount deposited will be used as pre-financing cost for marketing of NWFPs products • Amount realized as penalties shall be deposited in the group fund. • 5% of total products sold (revenue) shall be deposited to group account
Group fund management and utilization	• A joint account in favor of the group shall be opened under BDBL, Panbang. • Annual membership fees, fines and amount from sale proceeds shall be deposit in the group account. • The joint account will be managed by Executive Committee members. • Any expenditure incurred in the name of the group will be met from community development fund. • When the fund accumulates more than two hundred thousand, the fund will be utilized for community rituals, supporting the poor and disaster affected household. • If the group gets dissolved under unforeseen circumstances, the entire fund accumulated shall be equally divided among the members. • Welfare scheme will be decided by community member upon open discussion with the member
Conflict resolution	• Disputes amongst the members will be resolve by committee if not, • The conflicts that arise among the group members about their group activities and certain areas of NWFP harvesting will be put forward to Geog Tshogdu (GT) followed by Dzongkhag Tshogdu (DT) • If the conflict arises between NWFP group & outsider, case will be forwarded to DFO and Geog Administration by the committee

Meeting and purpose	In order to plan and discuss activities for the management of NWFPs, meeting will be conducted as follows; • General Assembly: All the members shall attend the meeting to discuss and plan activities before harvesting season; during April month every year. • Committee member: This meeting is mainly to discuss the planned activities as well as prepare progress reports (financial and physical) during the November month.
Penalties	• If the group members' violate the harvesting rules for NWFPs, he/she should pay the fine of Nu. 300/= and is not allowed to harvest the products in the next harvesting season • Any group members selling the products to outsider without notice of group will penalize fine of Nu. 2500/= • If outsider collects the products from the group's demarcated areas, they will be charging Nu. 5000/= as fine, if the product collected is for commercialization. • The owner of the cattle will be charges Nu.150/- cattle for damaging the NWFP collection area. • Member inactive after one year will be excluded from the group. • Income generated in group will be shared with the active member except inactive member will not be entitled.

Powers of the Department

The Department has the right to terminate the NWFP management and marketing group in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.

The deregistration, cancellation, termination and reinstatement of the NWFP plan shall be guided by Section 261 to 265 of the Forest and Nature Conservation Rules and Regulations, 2023.

Appendix I: Overview of Dendrocalamus hamiltonii production area

Overview of Dendrocalamus hamiltonii production area					
Block name	Sub-block	Total area (Ha)	Production area (%)	Production area (Ha)	
Wangling	Wangling	15	85	12.75	
Dangling 1	Danging 1	11.2	92	10.3	
Dangling 2	Dangling 2	8	96	7.68	
Kodong 1	Kodong1	4	98	3.92	
Kodong2	Kodong 2	3.8	88	3.34	
		42		37.99	

Appendix II: Block description form for Dendrocalamus hamiltonii

Wangling Block

Block description form for Dendrocalamus hamiltonii					
Date:					
Village:		Lamtang	Name of bamboo area:Wangling		
Block name:		Wangling	Total area (ha)		15
Access	What is the walking distance from the nearest village to the block ?				
	Less than 3 hours		Within a day		More than a day
	What is the walking distance from the block to the nearest road ?				
	Less than 3 hours		Within a day		More than a day
Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle <45%	
Forest Description	Dominant and associate species, canopy cover: Schima wallichii, castonopsis spp, Gmelia arborea				
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots
	1 Wangling		37	85	16
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None	
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable	
	1. Wangling	Yes			
Current management	Current management practices in entire block:				
Current threats	Threats	Description			
	Fire				
	Grazing	Need to control free cattle grazing			
	Illegal exploitation				
	Over-exploitation				
	Other:				
Opportunities	Future opportunities for the entire block: High production area if managed sustainably				

Dangling 1 Block

Block description form for Dendrocalamus hamiltonii							
Date:	31-01-24						
Village:	Lamtang			Name of bamboo area: Dangling1			
Block name:	Dangling 1			Total area (ha) 11.2			
Access	What is the walking distance from the nearest village to the block ?						
	Less than 3 hours		Within a day		More than a day		
	What is the walking distance from the block to the nearest road ?						
	Less than 3 hours		Within a day		More than a day		
Slope	Very steep >100%		Steep 70-100%		Moderate 45-70%		Gentle <45%
Forest Description	Dominant and associate species, canopy cover: Schima wallichii, castonopsis spp, Gmelia arborea						
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots		
	1 Wangling		27.6	92		11	
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None			
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable			
	1. Dangling1	Yes					
Current management	Current management practices in entire block: NWFP managed area						
Current threats	Threats	Description					
	Fire						
	Grazing						
	Illegal exploitation	Illegal collection by commuters					
	Over-exploitation						
	Other:						
Opportunities	Future opportunities for the entire block: High production area if managed sustainably						

Dangling 2 Block

Block description form for Dendrocalamus hamiltonii							
Date:	03-02-24						
Village:	Lamtang		Name of bamboo area:	Dangling2			
Block name:	Dangling 2		Total area (ha)	8			
Access	What is the walking distance from the nearest village to the block ?						
	Less than 3 hours		Within a day		More than a day		
	What is the walking distance from the block to the nearest road ?						
	Less than 3 hours		Within a day		More than a day		
Slope	Very steep >100%	Steep 70-100%		Moderate 45-70%	Gentle <45%		
Forest Description	Dominant and associate species, canopy cover: Schima wallichii, castonopsis spp, Gmelia arborea						
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots		
	1 Dangling 2		19.76	96	8		
Bamboo flower	Sporadic	Gregarious	Last time of mass flowering	None			
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable			
	1. Dangling2	Yes					
Current management	Current management practices in entire block: NWFP managed area						
Current threats	Threats	Description					
	Fire						
	Grazing						
	Illegal exploitation	Illegal collection by commuters					
	Over-exploitation						
	Other:						
Opportunities	Future opportunities for the entire block: Has future potential to increase the products through plantation.						

Kodong 1 Block

Block description form for Dendrocalamus hamiltonii					
Date:	04-02-24				
Village:	Lamtang	Name of bamboo area:	Kodong 2		
Block name:	Kodong 1	Total area (ha)	4		
Access	What is the walking distance from the nearest village to the block ?				
	Less than 3 hours		Within a day	More than a day	
	What is the walking distance from the block to the nearest road ?				
	Less than 3 hours		Within a day	More than a day	
Slope	Very steep >100%	Steep 70-100%		Moderate 45-70%	Gentle <45%
Forest Description	Dominant and associate species, canopy cover: Schima wallichii, castonopsis spp, Gmelia arborea, Daubanga				
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots
	1. Kodong 1		9.88	98	7
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None	
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable	
	1. Kodong 1	Yes			
Current management	Current management practices in entire block: NWFP managed area				
Current threats	Threats	Description			
	Fire				
	Grazing				
	Illegal exploitation	Illegal collection by commuters			
	Over-exploitation				
	Other:				
Opportunities	Future opportunities for the entire block: Has future potential to increase the products through plantation.				

Kodong 2 Block

Block description form for Dendrocalamus hamiltonii							
Date:	05-02-24						
Village:	Lamtang		Name of bamboo area:	Kodong 2			
Block name:	Kodong 2		Total area (ha)	3.8			
Access	What is the walking distance from the nearest village to the block ?						
	Less than 3 hours		Within a day		More than a day		
	What is the walking distance from the block to the nearest road ?						
	Less than 3 hours		Within a day		More than a day		
Slope	Very steep >100%	Steep 70-100%		Moderate 45-70%	Gentle <45%		
Forest Description	Dominant and associate species, canopy cover: Schima wallichii, castonopsis spp, Gmelia arborea, Daubanga grandefolia						
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots		
	1. Kodong 2		9.38	88	7		
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None			
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable			
	1. Kodong 2	Yes					
Current management	Current management practices in entire block: NWFP managed area						
Current threats	Threats	Description					
	Fire						
	Grazing						
	Illegal exploitation						
	Over-exploitation	Resources haevily collected					
	Other:						
Opportunities	Future opportunities for the entire block: Has future potential to increase the products through plantation.						

Appendix III: Resources assessment form for Dendrocalamus hamiltonii

Wangling Block

Date: 5/2/2024

Block: Wangling

Sample plot no: 1

Aspect: SE

Village: Lamtang

Local name of resource assessment area: Wangling

Slope: 55%

Slope correction Factor: 1.141

Altitude: 394m

Recorder: Kezang Dorji & Kiran Mongar

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	7	8	4	7	5	4	6	15
2	Good	8	7	6	5	5	4	5	14
3	Congested	17	10	6	9	9	8	9	26
4	Congested	11	12	10	8	11	8	9	28
5	Congested	5	9	12	10	7	9	6	22
6	Congested	5	6	5	6	5	4	6	15
7	Good	4	4	7	5	3	4	5	12
8	Congested	7	11	8	10	9	6	9	24
9	Congested	8	10	8	9	8	5	6	19
10	Damaged	4	3	1	1	2	1	1	4
11	Good	6	6	3	3	5	3	1	9
12	Good	5	5	6	6	3	6	5	14
		87	91	77	79	72	70	68	202

Dangling 2 Block

Date: 3/2/2024

Block: Dangling 2

Sample plot no: 3

Aspect: SE

Village: Lamtang

Local name of resource assessment area: Dangling 2

Slope: Less than 45%

Slope correction: 1.093

Altitude: 533m

Recorder: Kezang Dorji & Kiran Mongar

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	7	6	5	7	5	4	6	15
2	Good	7	8	6	6	6	5	5	16
3	Congested	11	12	11	9	9	10	7	26
4	Good	6	4	4	7	4	3	5	12
5	Good	5	5	6	5	5	5	6	16
6	Congested	9	11	10	9	9	9	8	26
7	Good	3	5	6	7	5	5	4	14
8	Good	5	6	4	6	5	5	4	14
		53	57	52	56	48	46	45	139

Kodong 1 Block

Date: 4/2/2024

Block: Kodong 1

Sample plot no: 4

Aspect: SW

Village: Lamtang

Local name of resource assessment area: kodong 1

Slope: 58%

Slope correction: 1.156

Altitude: 576m

Recorder: Sherab Dorji & Ugyen Dorji

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	4	5	6	7	4	5	6	15
2	Damaged	2	2	0	3	1	0	2	3
3	Damaged	1	3	2	0	2	1	0	3
4	Congested	8	6	8	5	5	8	3	16
5	Good	4	5	6	6	5	4	5	14
6	Good	5	7	6	5	5	6	5	16
7	Good	4	5	4	6	5	4	3	12
		28	33	32	32	24	28	24	79

Kodong 2 Block

Date: 5/2/2024

Village: Lamtang

Recorder: Kezang Dorji & Kiran Mongar

Block: Kodong 2

Local name of resource assessment area: Kodong 2

Sample plot no: 5

Slope: 52%

Slope correction: 1.127

Aspect: SW

Altitude: 403m

No of clump	Clump condition *	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	5	4	7	5	4	6	4	14
2	Good	4	7	7	5	7	5	4	16
3	Good	6	3	6	5	3	3	5	11
4	Damaged	2	2	4	2	2	1	1	4
5	Congested	9	8	9	6	7	7	6	20
6	Damaged	3	2	1	3	2	0	2	4
7	Good	6	5	5	7	5	3	6	14
		35	31	39	33	30	25	24	94

Appendix IV: Annual NWFP reporting form

Dzongkhag/Dungkhag:

Name of NWFP management and marketing group:

Gewog:

Villages:

Total number of Households:

NWFP Species	Quantity collected in a year	Qty sold in a year	Price received by the group per unit	Total income (Nu)	Description on resource availability	Remarks

Appendix V: List of NWFP Group Members

Sl. No.	Name	CID No.	Thram No.	House No.	Village
1	Chencho	12003000621	398	Wa-3-170	Lamtang
2	Pema Samten	12003000724	244	Wa-3-179	Lamtang
3	Pema Rinchen	12003000739	135	Wa-3-180	Lamthang
4	Phobzang	12003000716	76	Wa-3-178	Lamthang
5	Wangda	12003000758	181	Wa-3-181	Lamthang
6	Pema Wangchuk	12003001012	Nil	Wa-3-204	Lamthang
7	Dorji	12003000583	197	Wa-3-167	Lamthang
8	Sangay Lhendup	12003000925	232	Wa-3-194	Lamthang
9	Dorji	12003000806	195	Wa-3-185	Lamthang
10	Tshering	12003001002	669	Wa-3-203	Lamthang
11	Wangdhen	12003000663	128	Wa-3-259	Lamthang
12	Nima Dorji	12003000565	786	Wa-3-50	Lamthang
13	Sangay Loday	12003000947	Nil	Wa-3-196	Lamthang
14	Choida	12003000903	234	Wa-3-193	Lamthang
15	Karpola	12003001020	Nil	Wa-3-205	Lamthang
16	Pema Chogyel	12003000772	Nil	Wa-3-182	Lamthang
17	Sangay Juckchen	12003000671	240	Wa-3-173	Lamthang
18	Namgyel	12003000670	240	Wa-3-173	Lamthang
19	Sangay Karma	12003000778	Nil	Wa-3-182	Lamthang
20	Zangpo	12003000768	723	Wa-3-214	Lamthang